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Use of nanobubble water bioaugmented anaerobically digested sludge for high-efficacy energy production from high-solids anaerobic digestion of corn straw

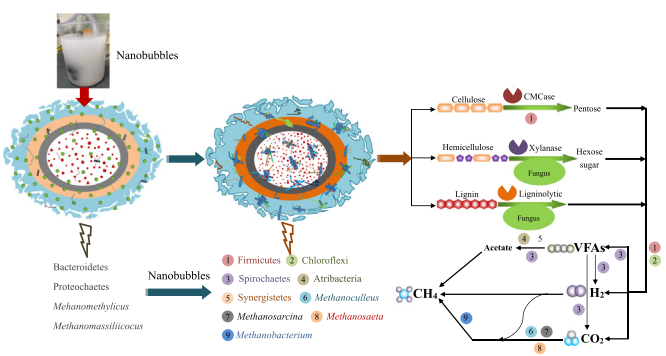
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HIGHLIGHTS

- NBW enhanced enzymatic activities and microbial community during bioaugmentation.
- NBW augmented sludge enhanced lignocellulose degradation in high-solids AD.
- 47% higher CH_4 yield was detected in O_2-NBW augmented sludge high-solids AD.
- No increase in reactor volume is realized in NBW augmented sludge high-solids AD.

GRAPHICAL ABSTRACT



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ABSTRACT

An increasing attention has been paid to the secure and sustainable management of agricultural wastes, especially lignocellulosic biomass. Nanobubble water (NBW) contains 10^6–10^8 bubbles/mL with diameter <1000 nm. Although previous studies have examined the enhancement effects of NBW on methane production from organic solid wastes, the NBW-based anaerobic digestion (AD) system is still restrained from practical application due to the large increase in AD reactor volume, generation of wastewater, and increase in energy consumption as well. In this study, NBW bioaugmentation of anaerobically digested sludge for the first time was performed for high-solids AD of corn straw. Results show that cellulase, xylanases and lignin peroxidase activities were increased by 2–55% during the NBW bioaugmentation process. Significant enrichment of hydrolytic/acidogenic bacteria and methanogenic archaea were noticed in the NBW bioaugmented sludge. This study clearly demonstrated 47% increase in methane production from high-solids AD of corn straw when O_2-NBW bioaugmented sludge was applied, achieving a net energy gain of 5138 MJ/t-volatile solids of corn straw with an energy recovery of 34%. The NBW-based high-solids AD system can provide a novel and sustainable management solution for renewable energy production from agricultural wastes, targeting the reduction of environmental pollution and energy crisis.

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1. Introduction

With the emergence of COVID-19 and post-COVID manufacturing industries, environmental pollution and energy crisis have become one of the most important socio-economic issues (Carbon Brief Ltd, 2021). The currently over reliance on fossil fuels has led to concerns on climate change and energy security. Therefore, new policies are urgently needed to reduce greenhouse gases (GHGs) emission, increase the recycling of waste biomaterials, and accelerate the development of circular economy. Anaerobic digestion (AD) has been widely applied for agricultural residues and organic wastes for decades to produce biogas that can further be used for electricity generation or thermal energy, thus reducing GHGs emission and fossil fuels' consumption.

Agricultural waste (mainly lignocellulosic biomass) is a promising and sustainable feedstock for the production of fuels, chemicals and other valuable products due to its abundance worldwide and enormous availability (Croce et al., 2016). However, the primary barrier hindering its widespread application is the low energy recovery from raw feedstocks via AD. Various pretreatments have been attempted to enhance the hydrolysis step, including physical (grinding, irradiation, and ultrasonic) (Zheng et al., 2014), chemical (acid and alkali) (Li et al., 2020), and biological (enzymatic, fungal) (Muaaz-Us-Salam et al., 2020) methods. However, in terms of energy balance, these pretreatment methods to a great extent make the whole system difficult to be compensated by the increased CH_4 yield. In addition, they also generate toxic by-products that are neither economical nor sustainable.

Development of sustainable energy production schemes is one of the most important scientific challenging subjects in the 21st century (Liu et al., 2010). As reported, nanobubble water (NBW) exhibits great potentials for the enhancement of bioenergy generation (Wang et al., 2021). NBW contains 10^6–10^8 bubbles/mL with diameter <1000 nm (Lyu et al., 2019). These nanobubbles (NBs) possess the following features including high specific surface area, hydrophobic property, high mass transfer rate, longevity, high mobility and diffusion of water molecules, capability to generate free radicals, etc. (Movahed and Sarmah, 2021). Therefore, NBs have been widely applied in various fields, including marine environments, water disinfection, membrane defouling, etc. (Lyu et al., 2019). Nowadays, much attention has been paid to the application of NBW in agricultural waste (lignocellulosic biomass) management due to its high mobility and diffusion of water molecules, zero pollution, and relatively low energy consumption. One of the most amazing properties is its improvement effect on the physiological activity of living organisms (Ebina et al., 2013). NBW has been proven to enhance methane production from waste activated sludge via AD (Wang et al., 2019). Yang et al. (2019) found that addition of NBW enhanced hydrolase activities in AD of waste activated sludge. NBW can also greatly enhance the hydrolysis-acidification stages of high loading AD systems (Wang et al., 2020b). Hou et al. (2021) reported that NBW can improve the coenzyme F_{420} activity at the methanogenesis stage in the two-stage AD of food waste. Most recently, O_2-NBW supplementation was proven capable of creating a more favorable microaerobic environment for facultative microbes to degrade organic solid waste via AD process (Wang et al., 2021; Hou et al., 2021), which has been successfully applied in wet AD systems for enhanced methane production from lignocellulosic biomass by 20–30% when compared to the control (Wang et al., 2020c). Nevertheless, the large volume increases in AD reactor and wastewater (liquid digestate) generation due to the direct addition of NBW lead to high operation costs and energy consumption, limiting the practical application of NBW in AD systems.

Bioaugmentation technology has been attempted to recover energy from ammonia-rich substrates to alleviate ammonia toxicity, achieving 31% increase in methane yield due to 5-fold increase in relative abundance of *Methanoculleus* sp. (Fotidis et al., 2014). Martin-Ryals et al. (2015) claimed that the continuous bioaugmentation of cellulolytic microbial consortia can improve the methanogenic performance. As reported, the methane yield can be increased by 10–22% from wet AD of corn straw when NBW pre-augmented anaerobically digested sludge was used (Wang et al., 2020a). However, the slightly increased CH_4 yield by the NBW-based wet AD technology to a great extent makes it difficult to compensate for the increases in energy consumption and biogas slurry production volume by the system due to NBW addition. In addition, the related mechanism of NBW bioaugmentation is still unclear, especially from the enzymatic and microbial levels in the pre-augmentation process.

High-solids AD (with total solids (TS) content ≥10%) is a promising alternative for organic solid waste management due to its high volumetric productivity, less heating energy consumption, much less or little liquid digestate production, and higher methane yields than wet AD. Up to the present, little information is available regarding the energy balance analysis of NBW-based AD systems, especially under high-solids conditions.

Therefore, in addition to the underlying mechanisms, this study aimed to figure out the effects of NBW-bioaugmented sludge on lignocellulosic degradation and methane production potential of corn straw under high-solids AD condition. The impact of NBW-bioaugmented sludge was firstly analyzed with respect to microbial communities and enzymatic activities. Then, the renewable energy production from high-solids AD of corn straw was evaluated in terms of methane yield. Finally, the major advantages of NBW-based high-solids AD technology were clarified based on energy gain analysis. The novel NBW-based high-solids AD system is expected to be more conducive to practical applications for energy production from lignocellulosic biomass, serving the sustainable society.

2. Materials and methods

2.1. Raw materials and anaerobically digested sludge

The ensiled corn straw was sampled from a local farm operated by Japan Agricultural Co-operatives in Ibaraki Prefecture, Japan. The corn straw was dried at 60 ± 1 °C to a constant weight and ground to pass through a 0.6 mm sieve, and then stored at room temperature before experiments. Anaerobically digested sludge was collected from the Shimodate Wastewater Treatment Plant in Chikusei, Ibaraki Prefecture, Japan. Before being used, the sludge was filtered through a 1-mm sieve and digested at 35 ± 1 °C for 7 days to activate microbial activity and decompose the remaining easily biodegradable organics. The main characteristics of corn straw and anaerobically digested sludge were described elsewhere (Wang et al., 2020a). The NBW used in this research was produced by a NBW generator (HACK, FB11, Japan). N_2 and O_2 (with a purity of 99.9995%) gases used for NBW generation were purchased from Taiyo Nippon Sanso Co., Ltd. (Japan). 1 L deionized water (DW) in a 5 L plastic beaker was injected into the NBW generator. During the NBW generation, the pressure was controlled at 0.22–0.25 MPa (25 ± 1 °C). The gas intake speed and DW circulation flow rate were controlled at 0.05 L/min and 3 L/min, respectively. After 20 min generation, the number of bubble particles was around 3×10^8 particles/mL with the particle size <200 nm. These data were similar with the characteristics of NBs obtained by Hou et al. (2022) and Wang et al. (2019).

2.2. Bioaugmentation of anaerobically digested sludge by deionized water, N_2-NBW, and O_2-NBW

In order to avoid the production of large amount of liquid digestate and increase in reactor volume caused by the direct NBW addition, use of NBW to augment the digested sludge first to enhance microbial community and enzymatic activity is regarded as a promising solution for efficient energy recovery from AD of corn straw. The bioaugmentation of anaerobically digested sludge was performed in 600 mL serum bottles under the addition of deionized water (DW), N_2-NBW or O_2-NBW. Each bottle with a working volume of 400 mL was added with 200 mL of anaerobically digested sludge and 200 mL of NBW or DW. Corn straw as the only substrate was added into the bottle with no addition of other substances like macro- and/or micronutrients. Addition of corn straw was controlled at a food-to-microorganisms (F/M, volatile solid (VS) basis) ratio of 1.5 at the beginning of each pre-augmentation. After being well mixed, the pH was adjusted to

7.00 ± 0.05 with 2.0 M NaOH or HCl. Then, the AD reactors were subjected to 5 min of N_2 flushing to create anaerobic conditions. All the pre-augmentation tests were performed in a thermostatic container at 35 ± 1 °C. The digestion residue was centrifuged at 5000 rpm for 10 min to remove the supernatant when methane production was <1% of the accumulated volume of methane during three consecutive days (Holliger et al., 2016). 2 mL of the digested sludge was sampled from each bioreactor during each bioaugmentation cycle (about 30 days) for the determinations of cellulase (CMCase), xylanase, and lignin peroxidase (LiP) activities. After the designated bioaugmentation cycles, the effluent from the AD reactor with DW or NBW addition was concentrated by centrifugation at 6000 rpm for 5 min to increase the slurry total solids (TS) content to 5–6%. Namely, equal amounts of DW or NBW and corn straw were added to repeat the above steps for 6 times, totally about 214 days for this pre-augmentation operation. All the experiments were conducted in triplicate. The concentrated AD effluent was used as inoculum for the high-solids AD process in Section 2.4.

2.3. Microbial community analysis

In order to identify the potential players in the biodegradation of corn straw during the augmentation process, the sludge samples were collected at the last bioaugmentation cycle for high-throughput 16S rRNA gene sequencing. The microbial DNA was extracted using the E.Z.N.A.® soil DNA Kit (Omega Bio-tek, Norcross, GA, US) according to the manufacturer's protocols. The primers 338F (5′-ACTCCTACGGGAG GCAGCAG-3′) and 806R (5′-GGACTACHVGGGTWTCTAAT-3′) were used to amplify the V3–V4 region of bacterial 16S rRNA (GeneAmp 9700, ABI, USA). The archaeal 16S rDNA was amplified with the primer pair of 524F10extF (5′-TGYCAGCCGCCGCGGTAA-3′) and Arch958RmodR (5′-YCCGGCGTTG AVTCCAATT-3′). According to the standard protocol of Majorbio Bio-Pharm Technology Co. Ltd. (Shanghai, China), the purified amplicons were combined on an Illumina MiSeq platform (Illumina, San Diego, USA) with equimolar and paired end sequencing (2 × 300 bp).

2.4. Biochemical methane potential (BMP) of corn straw via high-solids AD

High-solids AD of corn straw was performed in 100 mL closed Schott Duran serum bottles. In these bottles, the added mass weight of inoculum, i.e., the DW or NBW bioaugmented anaerobically digested sludge was 40 g, and the added mass weight of corn straw was estimated based on the F/M (VS basis) ratio of 3. The initial TS contents in all the reactors were 14–15% (high-solids AD). The bottles only loaded with the inoculum were prepared as blanks to eliminate the methane production amount from high-solids AD of the same dose inoculum only (without corn straw). Afterwards, all the treatment bottles were capped with rubber stoppers, flushed with N_2 gas for 5 min to create anaerobic condition, and placed in a temperature-controlled incubator (35 ± 1 °C). The reactors were manually shaken twice at fixed time points every day. Biogas volume was quantified by a 20 mL glass syringe connecting to the bottle, and biogas components were analyzed by a GC-8A/TCD (Shimadzu, Japan). The samples from each reactor were collected to analyze the cellulose, hemicellulose, and lignin contents, etc. All the sample analyses were conducted in duplicate.

2.5. Analytical methods for other parameters

TS and VS were measured by following the Standard Methods (APHA, 2012). pH was determined with a semi-solid pH meter (Testo 206, Germany). The contents of cellulose, hemicellulose, and lignin were quantified according to the procedures described elsewhere (Wang et al., 2020a). The activities of CMCase and xylanase were measured according to Weiβ et al. (2010). LiP activity was determined as the oxidation rate of veratryl alcohol by veratraldyde according to Pollegioni et al. (2015). The heating value (labeled herein as *E*) of corn straw during direct combustion was determined by a calorimeter (CA-4AJ, Shimadzu, Japan) after being dried at 60 °C till constant weight.

2.6. Energy balance and recovery

Energy balance analysis was conducted to evaluate the energy efficiency of the whole NBW-based high-solids AD system according to its simplified operation model as follows: The disposal of corn straw was assumed to be 1 ton VS of corn straw in this work. For the operation of the high-solids AD system, the input energy (E_{input}) for the digester was calculated as the energy required according to Eq. (1) modified from He et al. (2017).

$$E_{input} = mC(T_d - T_a) + 5\%mC(T_d - T_a) \quad (1)$$

where E_{input} (MJ/*t*) is the input energy for the high-solids AD process; *m* (*t*)is the digestate mass; *C* is the specific heat capacity of sludge, 4.18 MJ/*t*/°C; T_d is the high-solids AD temperature (35 °C in this study); and T_a is the ambient temperature (25 °C in this study). In addition, 5% of the heat consumption by the high-solids AD was estimated as the heat loss (Pilli et al., 2015).

Energy output (E_{output}, MJ/*t*) is the energy recovery from corn straw, which was calculated according to Eq. (2) based on methane production from biochemical methane potential (BMP) tests (through methane combustion).

$$E_{output} = q \times P_{CH_4} \times \zeta \quad (2)$$

where ζ is the lower heating value of methane (35.8 MJ/m^3); *q* (%) is the energy conversion efficiency (90%); and P_{CH4} (m^3 CH_4/*t*-VS) is the methane yield from the high-solids AD system.

ΔE is the energy balance or net energy gain calculated as the difference between the energy output and input (Eq. (3)).

$$\Delta E = E_{output} - E_{input} \quad (3)$$

The energy recovery (η) was calculated as Eq. (4) modified from a previous study (He et al., 2017).

$$\eta\ (\%) = \frac{\Delta E}{E} \times 100\% \quad (4)$$

where E is the heating value of corn straw during direct combustion.

2.7. Statistical analysis

One-way analysis of variance analysis was performed for statistical validation of the obtained results by using Origin 2019 software to estimate the significant difference among the treatments. A probability (*p* value) <0.05 was termed as significant. All data were calculated using Microsoft Office Excel 2020.

3. Results and discussion

3.1. Changes in performance of anaerobically digested sludge and its microorganisms during bioaugmentation

3.1.1. Methane production

In order to enhance the functions of microbes that can produce methane from AD of corn straw, NBW was added to augment the digested sludge for 6 cycles (214 days) with corn straw as the sole substrate in the AD process, and the DW-added reactor was used as the control. Fig. 1 shows the cumulative methane production during the augmentation process, which increased along with the proceeding of the augmentation cycles. In the first augmentation cycle, no significant difference ($p > 0.6$) was observed in the methane yields from the DW, N_2-NBW, and O_2-NBW reactors. The methane yields were gradually enhanced along with the augmentation cycles; at the end of the 6th augmentation cycle, the methane yields were increased by 20.0%, 29.7%, and 43.6% from the DW, N_2-NBW and O_2-NBW

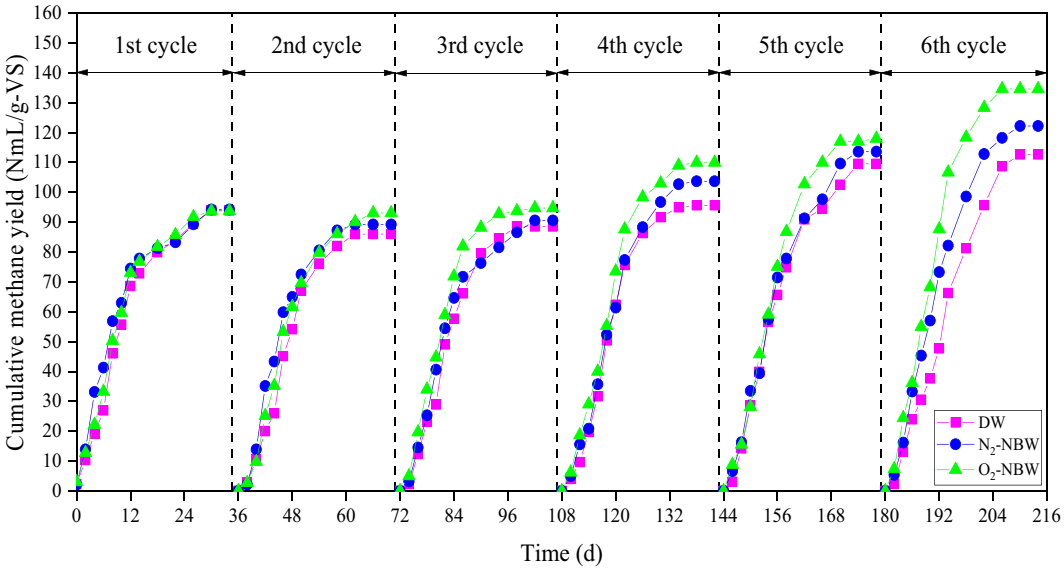


Fig. 1. Cumulative methane yields from anaerobic digestion of corn straw during 6 cycles of bioaugmentation with the addition of deionized water (DW) and nanobubble water (NBW), respectively.

reactors, respectively when compared to their 1st augmentation cycle. The O_2-NBW reactors obviously achieved the highest methane yield from AD of corn straw, about 20.3% higher than that from the DW reactors. Nguyen and Khanal (2018) claimed that microaerobic environment could enhance diversity and activity of facultative heterotrophs resulting in improved hydrolysis of substrates. As for the two kinds of NBWs, O_2-NBW showed better enhancement effect on methane yield than N_2-NBW, most probably due to the microaerobic environment created by O_2-NBW addition. In addition, O_2-NBs were observed to favor the generation of •OH radicals compared to N_2-NBs (Li et al., 2009). The oxygen-containing function groups can assist the degradation of lignocellulosic biomass (Yang et al., 2020). These previous works to some extent explained why the O_2-NBW reactors achieved higher methane production from AD of corn straw, most probably attributable to the facilitated facultative anaerobes, thus contributing to the enhanced hydrolysis stage in the AD process. The higher methane yield from lignocellulosic biomass may also be partially associated with the enhanced enzymatic activities and the augmented microorganisms that degrade lignocellulosic biomass.

3.1.2. Activities of related enzymes

CMCase can break down the β-(1,4)-glycosidic linkages and release glucose monomers (Karim et al., 2015), reflecting the degradability of cellulose in corn straw. Seen from Figs. 2a and b, after 6 cycles' augmentation, the CMCase activity was increased by 7.2%, 2.2%, and 2.5% in the DW, N_2-NBW, and O_2-NBW reactors, respectively. As for xylanase activity, the enhancement effect was more obviously, about 12.8%, 41.5%, and 54.6% increase in the DW, N_2-NBW and O_2-NBW reactors, respectively. Therefore, NBW addition could also be developed as a new strategy for AD systems to improve enzymatic activities. Among the test conditions, O_2-NBW supplementation might be the most efficient, with the highest CMCase and xylanase activities after the 6th bioaugmentation cycle, about 3.6% and 54.6% higher than those in the DW reactors, respectively. As Kholif et al. (2017) pointed out, the increased CMCase and xylanases activities may contribute to the improved methane production from AD of lignocellulose, which to some extent explained why methane yield was detected to increase in the DW or NBW bioaugmentation process. Seen from Fig. 2c, LiP activity was increased by 0.7%, 46.5%, and 4.3% in the DW, N_2-NBW, and O_2-NBW reactors after 214 days of augmentation, respectively. Lignin generally acts as a physical barrier hindering the accessibility of cellulose and hemicellulose to hydrolytic enzyme (Pang et al., 2021), leading to a low hydrolysis efficiency of corn straw. LiP can specifically oxidize the lignin polymers, thus enhancing its hydrolysis process which is the limiting step for AD of lignocellulosic biomass (Sánchez, 2009). The NBW-bioaugmented sludge could improve the hydrolysis of corn straw and then enhance the biodegradation of lignocellulosic biomass. Therefore, results from this work are very promising and could be potentially used for the development of an efficient high-solids AD process to manage lignocellulosic biomass for renewable energy production.

In the NBW reactors, the negatively charged surface of NBs may weaken the hydrogen bond network in the liquid phase (Chaplin, 2007). The adsorption of H^+ ions on the surface of NBs, and the movement or absence of protons in the liquid may trigger the modification or destruction of the hydrogen bond network (Chaplin, 2007). The weaken hydrogen bonds can cause the reduction in liquid viscosity (Ma et al., 2018), resulting in accelerated disintegration of lignocellulosic biomass and increased accessibility to the fermentative bacteria, which may further improve the hydrolysis efficiency of lignocellulosic biomass in the high-solids AD. In addition, the longer proton spin-spin relaxation time reflects the high mobility and diffusion of water molecules in NBW (Wang et al., 2020b), contributing to its high mass transfer efficiency with improved enzymatic activities (Alheshibri et al., 2016). The O_2-NBW reactors achieved the highest activities of hydrolytic enzymes in the AD of corn straw, possibly attributable to the highest metabolic activities of microbes owing to the improved electron transfer activity under the test conditions (Wang et al., 2021). After NBW bioaugmentation, the anaerobic microbes may keep their improved activities of hydrolytic enzymes and continue to exhibit high efficacy of degrading lignocellulose in the high-solids AD of corn straw. As known, variations of enzymatic activities in biological processes are associated with the shift of microbial communities. Therefore, the intrinsic mechanisms involved in the enhanced enzymatic activities and methane production from AD of corn straw by NBW-bioaugmented sludge deserve further elucidation in view of the changes in microbial communities.

3.1.3. Changes in microbial community structure

Microbial community composition is a major driving factor for the degradation of corn straw in AD, which is also an indicator of the activity of NBW-bioaugmented sludge. In order to better understand the role of NBW in the bioaugmentation process, microbial communities of the seed anaerobically digested sludge and three sludge samples from the DW, N_2-NBW, and O_2-NBW reactors after 214 days' augmentation were analyzed by high-throughput sequencing of 16S rRNA gene amplicons.

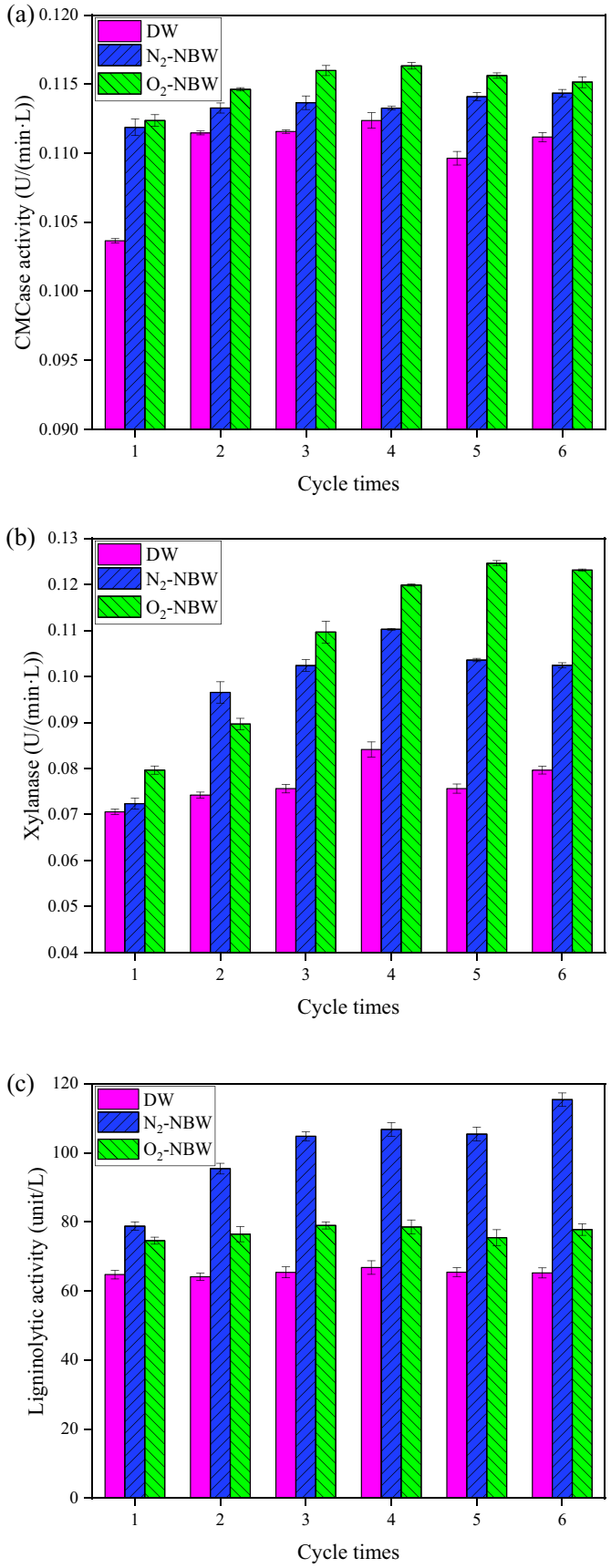


Fig. 2. Comparison of cellulase (CMCase), xylanase and ligninolytic activities during the bioaugmentation process with the addition of deionized water (DW) and nanobubble water (NBW), respectively.

As shown in Fig. 3a, a number of microbial groups associated with the biodegradation of corn straw were found, which were positively influenced by NBW. The relative abundances (RAs) of phyla Acidobacteria, Armatimonadetes, Atribacteria, Chloroflexi, Firmicutes, Spirochaetes, and Synergistetes were increased, while the RAs of Bacteroidetes and Proteobacteria were declined in the NBW bioaugmented reactors. Atribacteria belongs to heterotrophic anaerobes that can ferment propionate to produce acetic acid, thus resulting in enhanced methanogenesis (Hoshino et al., 2017). Chloroflexi can utilize and degrade carbohydrates at the initial stage of AD (hydrolysis and acidification) (Ma et al., 2019), which was more abundant in the NBW-bioaugmented reactors compared to the DW reactors. Although Firmicutes were detected in all the sludge samples, while the RAs of this phylum in the NBW reactors were higher than the DW reactors (6.2%, 15.1%, and 8.9% in the DW, N_2-NBW, and O_2-NBW reactors, respectively). However, without bioaugmentation, Firmicutes reached only 3.6% in AD of lignocellulosic biomass with O_2-NBW supplementation (Wang et al., 2020c). Firmicutes are claimed to produce extracellular enzymes (e.g., CMCase and proteases) and can work well under microaerobic condition to hydrolyze complex substrates (Fu et al., 2016). This may offer a reasonable explanation for the higher enzymatic activity (such as CMCase) in the NBW bioaugmentation reactors. In addition, Firmicutes can also utilize organics to produce volatile fatty acids (VFAs) and facilitate methane production. Spirochaetes are hydrolytic bacteria, which can convert carbon sources to VFAs and then to acetic acid, H_2 and CO_2 (Zhang et al., 2018). Synergistetes can well function in acetogenesis and accelerate the conversion of VFAs to acetate (Lee et al., 2009). As

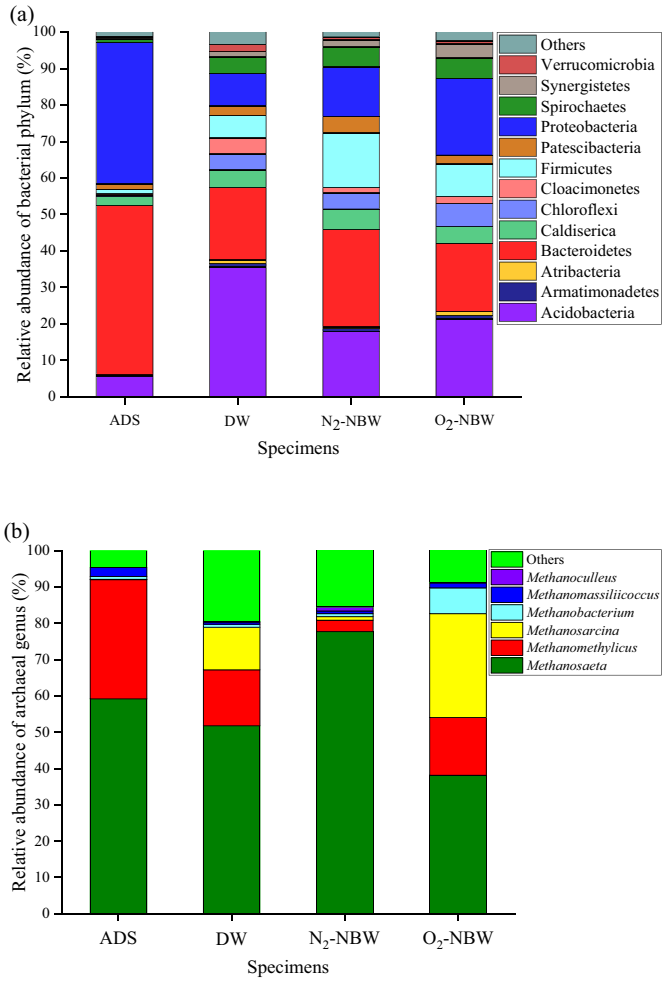


Fig. 3. Relative abundance of bacterial phylum (a) and archaeal genus (b). ADS, the seed anaerobically digested sludge; DW, bioaugmented sludge with deionized water; NBW, bioaugmented sludge with nanobubble water.

seen, these phyla are mainly hydrolytic bacteria, which can convert lignocellulosic biomass to small molecular intermediates (like VFAs), and further to acetic, H_2 and CO_2.

The phylogenetic classification of effective archaeal sequences was also summarized at genus level. The methanogen-rich inoculum was obtained from the NBW bioaugmentation process. Six genera including *Methanoculleus*, *Methanomassiliicoccus*, *Methanobacterium*, *Methanosarcina*, *Methanomethylicus*, and *Methanosaeta* were found to dominate all the sludge samples (Fig. 3b). In the N_2-NBW and O_2-NBW reactors, archaeal communities were also noticed to shift to a great extent. The RAs of *Methanoculleus* that can utilize H_2 and CO_2/formate as substrate to produce methane (Yuan et al., 2020) were 0.18%, 1.15%, and 0.22% after DW, N_2-NBW, and O_2-NBW bioaugmentation, respectively. The highest RA of *Methanobacterium* was detected in the O_2-NBW reactors, averagely reaching 7.1%, which belongs to hydrogen-consuming methanogens that can consume CO_2 and H_2 to produce methane (Yuan et al., 2020). Under no bioaugmentation condition, Wang et al. (2020b) noticed only 1% increase of *Methanobacterium* in AD of cellulose with NBW supplementation. Thus, NBW bioaugmentation may significantly enrich *Methanobacterium* sequences. As reported, *Methanobacterium* could be significantly enhanced under microaerobic conditions (Fu et al., 2016). These observations may explain why the highest RA of *Methanobacterium* was detected in the O_2-NBW bioaugmented sludge, probably due to that O_2-NBW may provide a proper microaerobic condition (because of O_2-NBs collapse to release trace O_2) that is beneficial for the growth of *Methanobacterium*. Wang et al. (2020c) verified that the microaerobic conditions provided by O_2-NBW exhibited positive effects on the methanogenesis stage of AD process. In addition, the RAs of *Methanosarcina* in the DW, N_2-NBW, and O_2-NBW reactors were 11.7%, 1.0%, and 28.6%, respectively, higher than that in the seed digested sludge. *Methanosarcina* has been reported to have direct interspecies electron transfer (DIET) with *Geobacter* (Chen et al., 2014). The establishment of DIET between anaerobic fermentation bacteria and methanogens can enhance AD of organic matters (Lü et al., 2016). *Methanosaeta* can also directly accept electrons via DIET from *Geobacter* and produce methane from CO_2 reduction (Rotaru et al., 2014). A lower RA of *Methanosarcina* was detected in the N_2-NBW reactors, attributable to the much greater proportion of *Methanosaeta*. The sum of RAs of *Methanosaeta* and *Methanosarcina* were 78.6% and 66.7% in the N_2-NBW and O_2-NBW reactors, about 23.9% and 5.1% higher than those in the DW reactors, respectively. Therefore, the increase in the sum of RAs of *Methanosaeta* and *Methanosarcina* in the NBW reactors may suggest the DIET occurrence due to NBW supplementation. This phenomenon agrees with the stimulation effect on microbial community by NBW supplementation (Wang et al., 2020c). These enhancements of microorganisms in the NBW reactors may imply the accelerated hydrolysis and methanogenesis stages of AD of corn straw when compared with the seed digested sludge.

In summary, the O_2-NBW reactors achieved the highest methane production (Fig. 1), most probably attributable to the microaerobic condition created by O_2-NBW. The high stability of O_2-NBs can continue to be the potent O_2 suppliers in the liquid (Ebina et al., 2013), which could provide O_2 into medium slowly and last for a longer time when compared with the ordinary O_2 aeration. In addition, the hydrophilicity and the high mobility of NBs can enhance the accessibility of microorganisms to corn straw in AD process, which could provide abundant nutrient and carbon sources for the growth of microorganisms, further enriching microbial communities. The addition of O_2-NBW may realize a delicate balance which not only facilitates the metabolism of facultative bacteria but also improves the enzymatic activities of microorganisms (Wang et al., 2021). NBW can generate numerous reactive oxygen species (ROS), including hydroxyl radicals (•OH), superoxide anion radicals (O_2^-), and singlet oxygen ($^1O_2$) during the process of NBs collapse. Shi et al. (2021) found the oxygen-containing function groups could facilitate the occurrence of DIET in AD process. Therefore, free radicals produced from NBs collapse may enhance the lignocellulosic biomass degradation. In the previous works, NBs in NBW were claimed to almost disappear and no large bubbles were observed after storage at room temperature for one month (Wang et al., 2020c; Yang et al., 2019; Wang et al., 2019). Therefore, it's hypothesized that the NBs collapse occurs and disappears in the liquid. However, the stable existence time of NBs still remains a big challenge. Currently the NBW studies are at the preliminary stage, and the disappearance mechanism of NBs in liquid needs to be further explored by using the innovative technologies. Still, some direct evidence is demanding for the above viewpoints in the followed-up works.

3.2. Use of bioaugmented sludge for BMP tests on high-solids AD of corn straw

3.2.1. Degradation of cellulose, hemicellulose, and lignin

During the high-solids AD of corn straw, the three major organic components were found to decrease obviously, with the largest reduction being detected in hemicellulose (Fig. 4). As seen, the contents of cellulose, hemicellulose, and lignin in the NBW reactors were averagely reduced to 10.9–13.6%, 7.5–8.8%, and 6.3–6.4% ($p < 0.05$), respectively in comparison to 16.2%, 8.5%, and 7.3% in the DW reactors. These results suggest that the NBW-bioaugmented sludge exhibited higher degradation capabilities for lignin and cellulose. In addition, the O_2-NBW bioaugmented sludge seemed to have the highest lignocellulosic biomass degradation capability than the N_2-NBW and DW counterparts, with cellulose, hemicellulose, and lignin reduced by about 50.4%, 60.0%, and 31.7%, respectively ($p < 0.05$). This observation suggests that the degradation capability of lignocellulose in corn straw follows a descending order of hemicellulose > cellulose > lignin in the high-solids AD. Seen from Section 3.1.3, a significant enrichment of hydrolytic/acidogenic bacteria and methanogenic archaea was noticed in the O_2-NBW bioaugmented sludge compared to the N_2-NBW reactors. Therefore, the O_2-NBW bioaugmented sludge also possessed the highest total lignocellulose (cellulose + hemicellulose + lignin) degradation potential, about 51% reduction in lignocellulose content in corn straw, followed by the N_2-NBW (44%) and DW (36%) bioaugmented sludge, respectively. NBW-based high-solids AD technology was found to be able to efficiently hydrolyze cellulose, hemicellulose, and lignin, resulting in high efficiency methane production and energy recovery from corn straw.

3.2.2. Methane production

Methane production from high-solids AD of corn straw by using the bioaugmented sludge as inoculum was recorded (Fig. 5). It was noted that almost no biogas production from the blank reactors (only the inoculum, or the bioaugmented sludge with DW, N_2-NBW or O_2-NBW addition). During the first eight days, little methane production was detected from all the

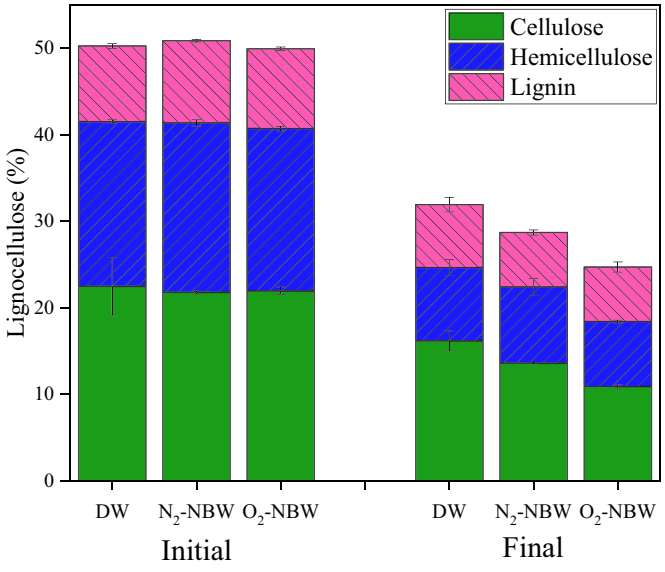


Fig. 4. Variations of cellulose, hemicellulose, and lignin contents before (initial) and after (Final) high-solids anaerobic digestion of corn straw by anaerobically digested sludge bioaugmented with the addition of deionized water (DW), N_2-NBW, and O_2-NBW. NBW, nanobubble water.

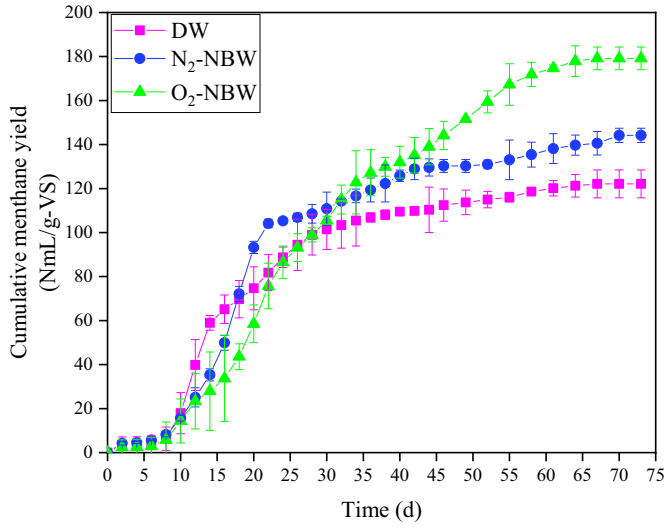


Fig. 5. Cumulative methane yields from high-solids anaerobic digestion of corn straw by using anaerobically digested sludge bioaugmented with the addition of deionized water (DW), N_2-NBW, and O_2-NBW. NBW, nanobubble water.

high-solids AD reactors. Afterwards, the methane production was observed to increase remarkably till day 64 and then became level off. During the initial stage (days 0–16), the DW-bioaugmented sludge reactors produced the highest methane volume, most probably due to the fact that the NBW bioaugmented sludge reactors were predominated by hydrolysis/acidification process. As discussed in Section 3.1.3, the NBW-bioaugmented sludge reactors possess higher hydrolysis capacity of lignocellulosic biomass due to the enriched hydrolytic bacteria. Therefore, these reactors may produce more VFAs, leading to a lower local pH microenvironment, which may inhibit the rapid growth of methanogens. Further in-depth studies are still necessary on the activity of methanogens during this early stage of high-solids AD, and the possible solutions or strategies for this delayed methane production as well. More specifically, during days 0–28, the lowest methane yields were obtained from the O_2-NBW-bioaugmented sludge reactors. This observation may be more associated with the micro-oxygen environment favoring the enrichment of facultative bacteria to degrade corn straw (Fu et al., 2016), which might result in the lowest pH microenvironment (data not shown). When the small molecular intermediates were gradually consumed, the local pH microenvironment would return to neutral along with the operation, resulting in higher methane production during the later stage.

As seen, during the 70 days' BMP tests, the cumulative methane yields from corn straw in the N_2-NBW and O_2-NBW reactors reached 144.2 ± 3.2 ($p = 0.048 < 0.05$) and 179.2 ± 5.2 ($p = 0.010 < 0.05$) NmL/g-VS, increasing by 18.1% and 46.8% when compared to those from the DW reactors (averagely 122.1 ± 6.3 NmL/g-VS). Compared with the result from He et al. (2022) who achieved 12–25% increase in methane yields from AD of corn straw with H_2-NBW addition, the enhancement effect on methane production from corn straw after NBW bioaugmentation is more prominent. In a previous study, the theoretical methane production of corn straw was 223.3 mL/g-VS according to the elemental composition (C, H, N and O) of corn straw (Wang et al., 2020a). The highest biodegradability of corn straw was achieved in the O_2-NBW pre-augmented sludge reactors, about 80%, followed by N_2-NBW (65%), and DW (55%). This enhancement effect on methane production was mainly attributable to the enriched microbial communities and enhanced enzymatic activities in this study. Up to today, the O_2-NBW-based high-solids AD system achieved the highest methane production from corn straw among all the test NBW-based AD systems.

3.2.3. Energy balance analysis

Although the NBW-bioaugmented sludge reactors achieved some increase in methane production during the 70 days' high-solids AD of corn straw, this increment of energy (methane) may be not enough to support the energy consumption by maintaining the operation of AD reactors. In this study, the energy balance was analyzed based on the BMP tests on high-solids AD of corn straw according to Eqs. (1)–(4) (Table 1). The heating value (E) of corn straw was determined as 15,160 ± 25 MJ/t-$VS_{corn\ straw}$. In fact, a positive net energy gain was achieved in all the reactors. It is worth noting that the O_2-NBW bioaugmented sludge reactors achieved the highest net energy gain (ΔE = 5138.0 MJ/t-$VS_{corn\ straw}$) and energy recovery (η = 33.9%) through methane production from high-solids AD of corn straw, followed by N_2-NBW (ΔE = 4010.3 MJ/t-$VS_{corn\ straw}$, 26.5%) and DW (ΔE = 3298.3 MJ/t-$VS_{corn\ straw}$, 21.8%) bioaugmented sludge reactors, respectively. These results clearly highlight the much greater net energy gain from the high-solids AD of corn straw when the O_2-NBW bioaugmented sludge was applied, increasing by 55.8% in net energy gain and 55.5% in energy recovery respectively when compared to the control (DW bioaugmented sludge reactors). This is mainly attributable to the much higher methane production (about 47% increase) from the O_2-NBW bioaugmented sludge reactors (Table 1 and Fig. 5).

3.3. Implications of NBW-based high-solids AD systems

Improper handling (like incineration) or disposal of corn straw would cause negative effects on human health and the environment, which generally occurs in most developing countries. NBW-based high-solids AD can not only achieve sustainable management of lignocellulosic biomass but also obtain high net energy gain from lignocellulosic biomass. NBW technology has been frequently trialed on AD of organic solid wastes due to its unique characteristics and properties including nontoxicity and low energy consumption that could avoid secondary pollution and extra energy consumption, targeting the sustainable development goals. The new concept proposed in this study, including NBW bioaugmentation of anaerobically digested sludge and subsequent high-solids AD of lignocellulosic biomass, can avoid the increase in reactor volume and the generation of liquid digestate (wastewater), greatly improving the volumetric methane productivity and thus lowering the investment and operation costs. This study first confirmed that NBW bioaugmented AD sludge could achieve enhanced methane yield and net energy gain from high-solids AD of corn straw, clearly demonstrating the economic feasibility of the NBW-based high-solids AD system for organic solid wastes. The apparent enhancement roles of enzymatic activities and microbial communities are interesting from both scientific and practical viewpoints, which may serve as potential indicators for a well and stably operated high-solids AD reactor.

4. Conclusions

NBW bioaugmentation can obviously enhance the enzymatic activities (2–55%) and microbial communities (7–9%) in anaerobically digested sludge, especially lignocellulolytic microorganisms. Cellulose, hemicellulose, and lignin reductions can also be enhanced in the NBW-based high-solids AD systems. Results from this work can help better understand the fundamentals involved. In this study, it was first confirmed that the NBW-based high-solids AD technology could increase net energy gain by 56% from corn straw when compared to the control. NBW-based high-solids

Table 1
Energy balance analysis and energy recovery from high-solids anaerobic digestion of 1 ton volatile solid of corn straw.

Parameters	Reactors with pre-augmented sludge		
	DW	N_2-NBW	O_2-NBW
Energy input (E_{input}, MJ/t)	635.8		
Methane yield (NmL/g-VS)	122.1	144.2	179.2
Energy output (E_{output}, MJ/t)	3934.1	4646.1	5773.8
Net energy gain (ΔE, MJ/t)	3298.3	4010.3	5138.0
Heating value of corn straw (E, MJ/t)	15160 ± 25		
Energy recovery (η, %)	21.8	26.5	33.9

AD technology is expected to provide a cost-effective promotion approach for the degradation of lignocellulose via a facile and green means. This technology can improve methane production with no chemical addition and no additional energy consumptions, which is consistent with the sustainable development goals.

CRediT authorship contribution statement

Xuezhi Wang: Writing – original draft, Conceptualization. **Zhongfang Lei:** Writing – review & editing, Supervision. **Zhenya Zhang:** Writing – review & editing. **Kazuya Shimizu:** Writing – review & editing. **Duu-Jong Lee:** Writing – review & editing. **Samir Kumar Khanal:** Writing – review & editing.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Corrigendum



Corrigendum to "Use of nanobubble water bioaugmented anaerobically digested sludge for high-efficacy energy production from high-solids anaerobic digestion of corn straw" [Sci. Total Environ. 863 (2023) 160825]

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This Corrigendum relates to the above article.

In this article, the authors used wording that overstated the implications of the experimental results with respect to sustainability and economic feasibility. The study reported enhanced methane production and a positive process-level net energy balance under the laboratory conditions tested.

However, the study did not include a lifecycle assessment, techno-economic analysis, pilot-scale validation, or full environmental sustainability assessment. The authors therefore wish to clarify that the results should be interpreted as evidence of process-level energy performance, rather than as a demonstration of environmental sustainability or economic feasibility.

The original version of Section 3.3 on page 7 is as below:

NBW-based high-solids AD can not only achieve sustainable management of lignocellulosic biomass but also obtain high net energy gain from lignocellulosic biomass. This study first confirmed that NBW bioaugmented AD sludge could achieve enhanced methane yield and net energy gain from high-solids AD of corn straw, clearly demonstrating the economic feasibility of the NBW-based high-solids AD system for organic solid wastes.

The updated version should be as below:

The present study shows that nanobubble water bioaugmented anaerobically digested sludge can enhance methane production from corn straw and generate a positive process-level net energy balance under the tested laboratory conditions. These findings indicate the potential of NBW-assisted high-solids anaerobic digestion for lignocellulosic biomass valorization and renewable energy recovery. However, the energy-balance calculation presented here should be interpreted as preliminary and process-specific and should not be taken as evidence of full environmental sustainability or economic feasibility. Further lifecycle assessment, techno-economic analysis, and pilot- or full-scale validation are required before conclusions can be drawn on the broader sustainability, economic viability, and practical implementation of this approach.

The authors would like to apologise for any inconvenience caused.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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